TEMASEK JUNIOR COLLEGE

Preliminary Examinations 2013

BIOLOGY	**8875/01**
Higher 1	**Thursday, 26 September**
Paper 1 Multiple Choice	**1 hour**

Additional Materials: **Optical Mark Sheet (OMS)**

READ THESE INSTRUCTIONS FIRST

Write in a soft pencil.
Do not use staples, paper clips, highlighters, glue or correction fluid.
Write your name, Centre number and index number on the Answer Sheet in the spaces provided.

There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.
Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

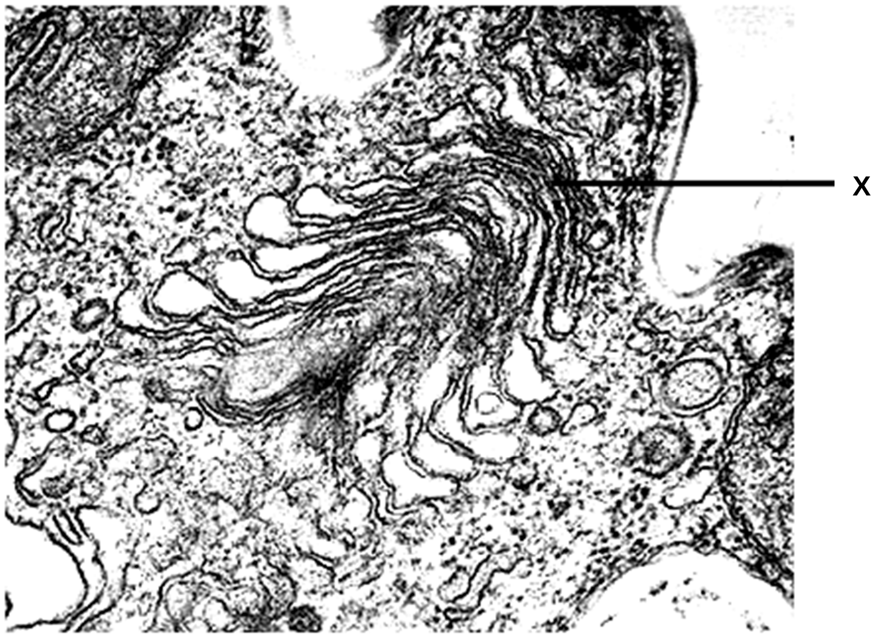
Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for the wrong answer.
Any rough working should be done in this booklet.
Calculators may be used.

This document consists of **21** printed pages and **1** blank page.

[Turn over

1 Which of the following are the most likely consequences for a cell lacking structure **X**?



1 The cell dies because it is unable to make glycoproteins to detect stimuli from its environment.
2 The cell dies from a lack of enzymes to digest food taken in by endocytosis.
3 The cell dies from the accumulation of worn out organelles within itself.
4 The cell is unable to reproduce itself.
5 The cell is unable to export its enzymes or peptide hormones.

A 2 and 3

B 2 and 5

C 3, 4 and 5

D 1, 2, 3 and 5

2 A microscope is set up to examine the surface of a leaf. The 40x objective lens and a 7x eyepiece is being used. The diameter of the field of view is 0.4mm.

What is the approximate density of the stomata if 15 of them are visible under these conditions?

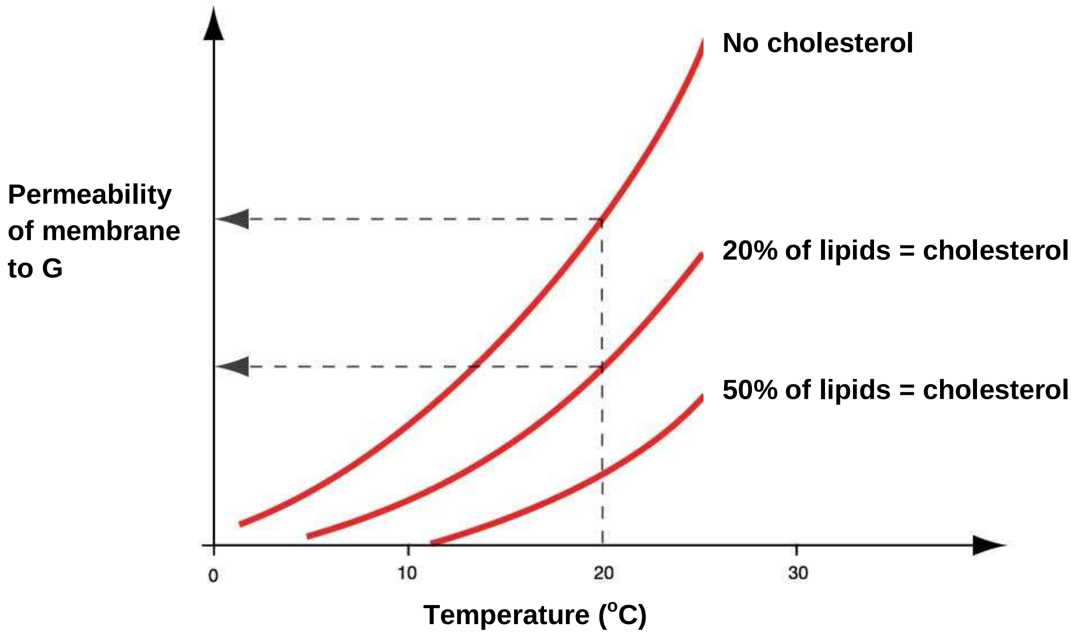
A 33 stomata/mm^2

B 120 stomata/mm^2

C 528 stomata/mm^2

D 1680 stomata/mm^2

3 The graph below shows the permeability of three different membranes to chemical **G** at different temperatures. These three membranes differ in the amount of cholesterol present in the phospholipid bilayers.



Which of the following is a possible explanation for the observed data?

A Increase in temperature increases the permeability of the membrane to **G** as cholesterol increases the fluidity of the phospholipids.

B At 20°C, an increase in the proportion of cholesterol in the membrane increases the permeability of membrane to **G** as both cholesterol and **G** are non-polar.

C Increase in the proportion of cholesterol decreases the permeability of the membrane to **G** as cholesterol decreases the fluidity of the phospholipids.

D With an increase in the proportion of cholesterol in the membrane, a lower temperature is required to achieve the same level of permeability for **G** as **G** will gain a higher kinetic energy to penetrate the membrane.

4 Which of the following are true of triglycerides and phospholipids?

1 Both contain glycerol.
2 Both are amphipathic.
3 Triglycerides have higher carbon content than phospholipids for storage purposes.
4 Phospholipids have hydrophilic regions to interact with cell cytoplasm unlike triglycerides.
5 Triglycerides are formed from glycerol and fatty acids but phospholipids are formed from glycerol and phosphoric acid.

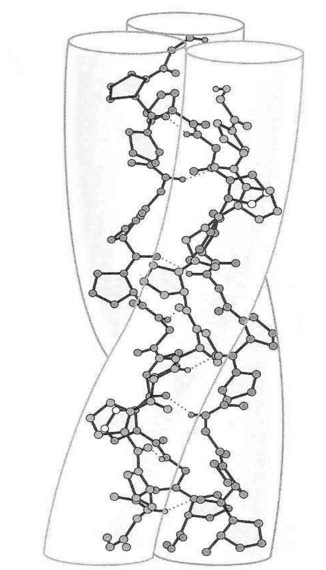
A 1, 3 and 5

B 1, 2 and 3

C 1, 3, and 4

D 2, ,3 and 4

5 The molecule shown below is found in various parts of the human body, and is helical in structure.



Which of the following statement about this molecule is true?

A Complementary base pairing occurs through the formation of hydrogen bonds between chains.

B Hydrogen bonds within each chain give each chain its helical structure.

C Glutamine occurs very frequently in the sequence of the subunits in the molecule.

D The entire molecule interacts with proteins to form complexes necessary for the control of cell activities.

6 The enzyme amylase hydrolyses starch into disaccharides. Amylase is unable to break down cellulose because

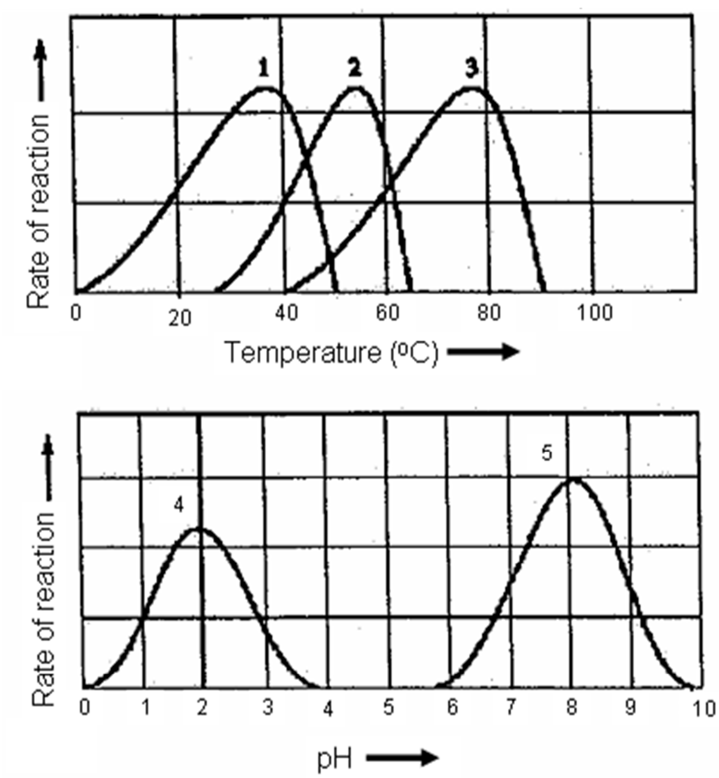
A cellulose does not contain glycosidic bonds.

B starch consists of glucose; cellulose consists of other monosaccharides.

C the bonds between sugars in cellulose are much stronger.

D the sugars in cellulose are bonded together differently than in starch, hence cellulose have a different shape.

7 The following graphs show the activities of different enzymes (1 to 5) under different conditions.



Which statement is a correct explanation for the rate of reaction of different enzymes?

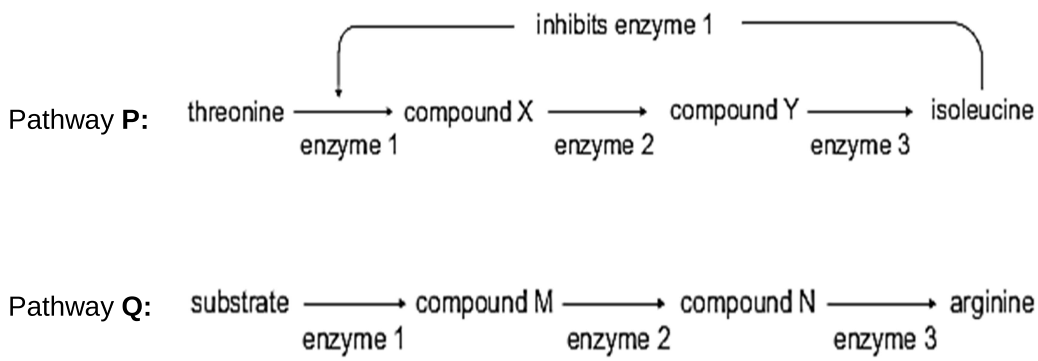
A Kinetic energy of enzyme 3 and its substrate is highest at 75°C.

B At pH 2, most of the R groups at the active site of enzyme 4 are all negatively charged.

C At pH 8.1, substrate is bonded to the active site of enzyme 5 by hydrogen bonds only.

D At 60°C, several hydrogen bonds between R groups of enzyme 2 are broken.

8 In the production of isoleucine from threonine (pathway **P**), the end product acts as an inhibitor of the first enzyme in the pathway. In the production of arginine (pathway **Q**), the end product has no influence on other enzymes in the pathway. The two pathways are shown in the figure.



Which of the following conclusions are correct?

1. In pathway **P**, if the production of enzyme 3 stops, there would be increased production of isoleucine.
2. In pathway **Q**, if the production of enzyme 3 stops, there would be decreased production of arginine.
3. In pathway **P**, provided all enzymes are present, the production of isoleucine would be continuous if there was a continuous supply of threonine.
4. In pathway **Q**, provided all enzymes are present, the production of arginine would be continuous if there was a continuous supply of substrate.

A 1 and 3 only

B 1 and 4 only

C 2 and 3 only

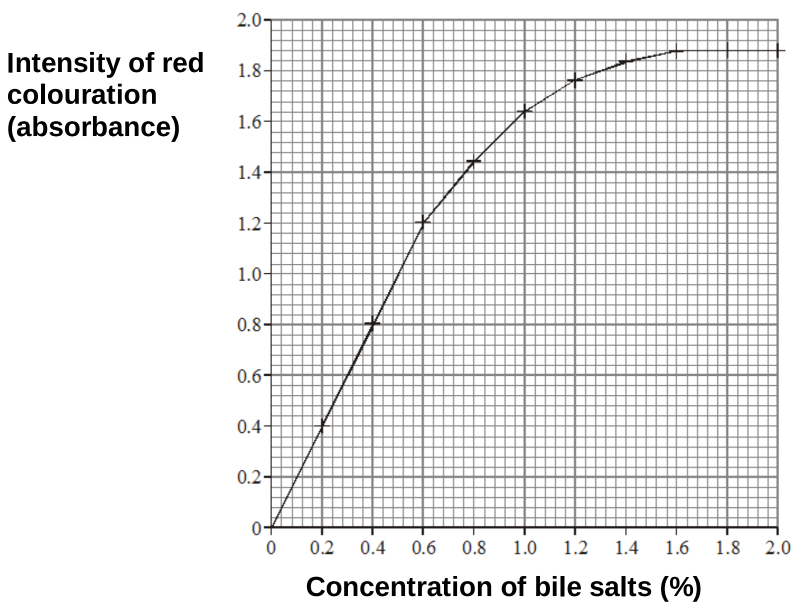
D 2 and 4 only

9 Beetroots are root vegetables. They appear red because their cells contain a water soluble red pigment in their vacuoles, which cannot pass through the membrane surrounding the vacuole (tonoplast).

Five beetroot discs were then placed in a tube containing 25 cm^3 of 2% bile salt solution and left for 30 minutes at 20℃. The proce dure was repeated for different concentrations of bile salts and one set of discs was left in distilled water.

After 30 minutes, each set of beetroot discs was removed from the solutions and from the water. Each bile salt solution was stirred and a sample removed and placed in a colorimeter. The intensity of red coloration (absorbance) was determined by the colorimeter.

The results of the investigation are shown in the graph below.



What is the best explanation for the results?

A Pigments leaked out via protein channels in the plasma membrane.

B Bile salts caused phospholipids in both tonoplast and plasma membrane to be emulsified.

C Bile salts did not have any effect on the tonoplast.

D Temperature affected the intensity of colour absorbance.

10 A diploid cell contains four chromosomes (2n = 4). If this cell divides by mitosis, and there is no mutation, how many genotype(s) exist among the daughter cells?

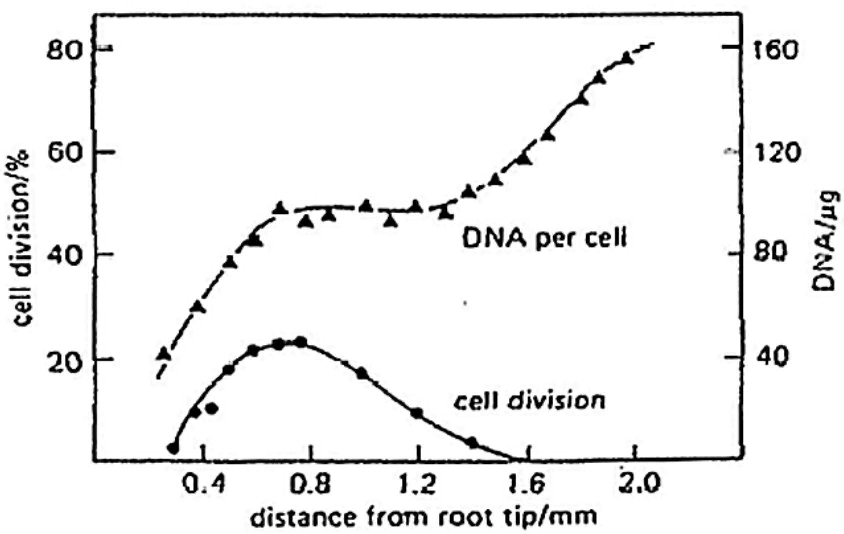
A 1

B 2

C 4

D 8

11 The graph below shows the changes in DNA content and rate of division in cells at varying distances from the tip of an onion root.



Which one of the following can be determined from the graph?

A Cells nearest the tip are dividing by meiosis.

B DNA quantity varies inversely with the cell division rate.

C DNA synthesis precedes cell division.

D The cells become polyploidy.

12 The table below shows a list of characteristics displayed by mutant strains of *E.coli* during DNA replication and the possible reasons.

No	Characteristics	Enzymes or functions affected by mutation
1	Okazaki fragments accumulate and DNA synthesis is never completed	DNA ligase activity is missing
2	Supercoils are found to remain at the flanks of the replication bubbles	DNA helicase is hyperactive
3	Synthesis of new strand is very slow	Substrate binding site of RNA polymerase slightly altered
4	No initiation of replication occurs	A-T rich region at origin of replication deleted

Which of the reasons correctly explain the characteristics displayed by the mutant *E. coli* strains?

A 1 and 2
B 1 and 3
C 1, 3, and 4
D All of the above

13 In three different genetic dictionaries, the genetic code for the amino acid cysteine is given as:

I **ACA** or **ACG**
II **TGT** or **TGC**
III **UGU** or **UGC**

The explanation for this may be:

1 Some genetic dictionaries show mRNA codons, others show DNA triplets.

2 The genetic code can be read in either the 3' or 5' direction along the DNA.

3 Some genetic dictionaries show the triplet code on the DNA strand complementary to the mRNA code, others show the triplet code on the other DNA strand.

4 The genetic code is a degenerate triplet code.

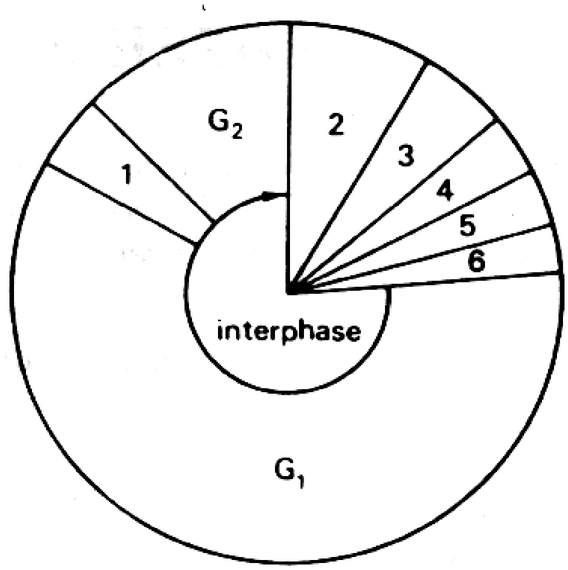
A 3 only

B 2 and 4 only

C 1, 2 and 3 only

D 1, 3 and 4 only

14 The diagram below shows the cell cycle. G_1 and G_2 represent periods of intensive synthesis within the cytoplasm.

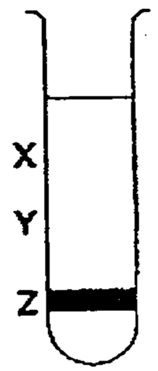


Which of the following correctly identifies the other cellular events associated with mitosis?

	1	2	3	4	5	6
A	cytokinesis	DNA synthesis	prophase	telophase	metaphase	anaphase
B	telophase	cytokinesis	DNA synthesis	prophase	metaphase	anaphase
C	DNA synthesis	prophase	metaphase	anaphase	telophase	cytokinesis
D	prophase	metaphase	anaphase	telophase	DNA synthesis	cytokinesis

15 A culture of bacteria had all its DNA labelled with the heavy isotope of nitrogen, ^{15}N. The bacterial culture was then allowed to reproduce using nucleotides containing normal ^{14}N. The DNA was examined using a centrifuge after one generation and again after two generations.

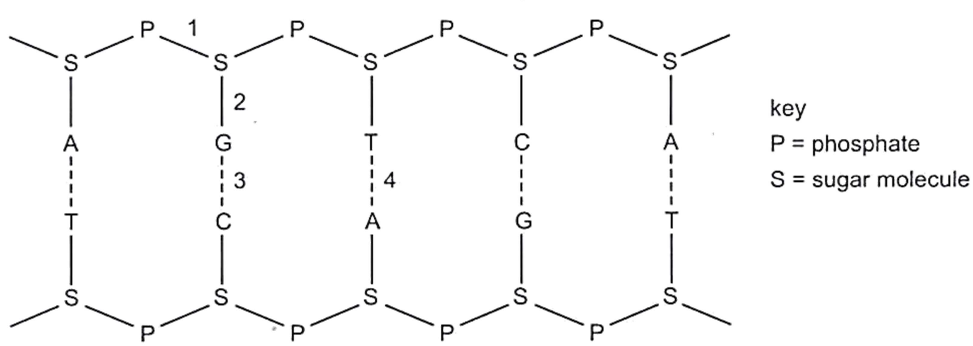
The diagram shows the position of the DNA band at **Z** in the centrifuge tube when the DNA was first labelled.



Where would the DNA be found after the second and after the third cell generations?

	After second generation	After third generation
A	Half at **X** and half at **Y**	One quarter at **X** and three quarter at **Y**
B	Half at **X** and half at **Y**	One quarter at **Y** and three quarter at **X**
C	One quarter at **X** and three quarter at **Y**	Half at **X** and half at **Y**
D	One quarter at **Y** and three quarter at **X**	Half at **X** and half at **Y**

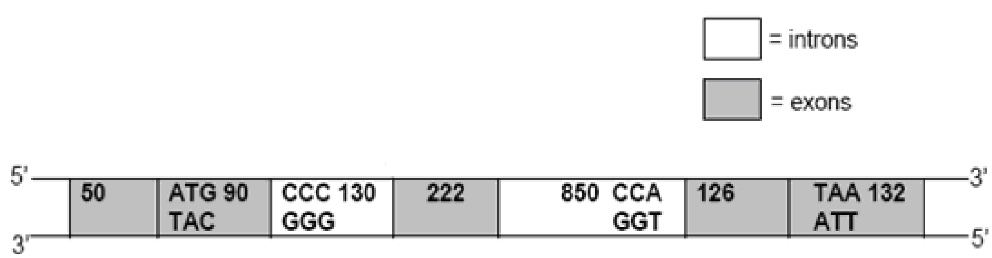
16 The diagram shows part of a nucleic acid.



Which row correctly describes the bonds shown in the diagram at positions 1, 2, 3 and 4?

	Is formed by condensation	**Forms a di-ester**	**Occurs during transcription**	**Involves attraction between polar molecules**
A	1	2	1, 2 and 3	3
B	1 and 2	1	3 and 4	3 and 4
C	2, 3 and 4	1	3 and 4	1, 3 and 4
D	2	3	1, 3 and 4	4

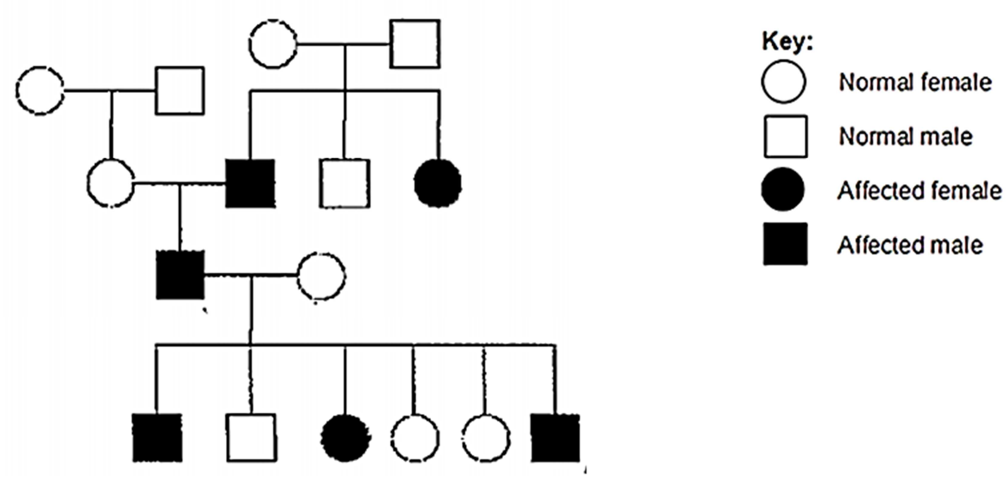
17 The diagram below shows the genomic structure of a human gene. The numbers within the boxes indicate the length of nucleotides of each region, inclusive of the bases stated in the diagram. The DNA sequences corresponding to the start codon and the stop codon are also indicated.



What is the length (in nucleotides) of the primary RNA transcript (pre-mRNA) and how many amino acids are present in the protein?

	Length of primary RNA transcript	**Number of amino acid in protein**
A	1 600	146
B	620	146
C	1 600	206
D	620	206

18 Based on the pedigree chart below, what is the most likely inheritance pattern?



A Autosomal dominant

B Autosomal recessive

C X-linked dominant

D X-linked recessive

19 In lentils, the seed coat pattern is determined by a gene with 3 alleles. The phenotypes are marbled, spotted and clear. 4 crosses were repeated many times. The crosses and the outcomes of these crosses are shown in the table below.

Cross	Parents	Offspring phenotype and ratio
1	marbled x marbled	3 marbled : 1 clear
2	spotted x clear	1 spotted : 1 clear
3	marbled x marbled	3 marbled : 1 spotted
4	clear x clear	all clear

From the data, it is possible to conclude that

A spotted and clear alleles are co-dominant.

B clear offspring can be either homozygous or heterozygous.

C two-thirds of the marbled offspring in cross 3 are heterozygous.

D the spotted and clear parents in cross 2 are homozygous for the spotted and clear alleles respectively.

20 The genotype of F_1 individuals is **AaBbCcDd**. Assuming independent assortment of these four genes, what is the probability that the F_2 offspring would express the phenotype of genotype **aabbccdd**?

A 1/4

B 1/16

C 1/256

D 1/512

21 When organochlorine insecticides such as DDT were in widespread use, mosquitoes in malarial regions developed resistance more rapidly than houseflies in Britain.

What could account for the difference in the rates of the development of resistance?

A More insecticides were used in Britain.

B More insecticides were used in malarial regions.

C Mosquitoes have fewer random mutations when exposed to insecticides.

D Mosquitoes have more random mutations when exposed to insecticides.

22 The wounds of patients who have undergone surgery are susceptible to bacterial infections by *Enterococcus faecium* and methicillin-resistant *Staphylococcus aureus* (MRSA). Methicillin and other antibiotics are often used to treat infections of surgical wounds, but do not always cure them.

Which of the following statements about the development of resistance to antibiotics in bacteria such as *Enterococcus faecium* and MRSA is correct?

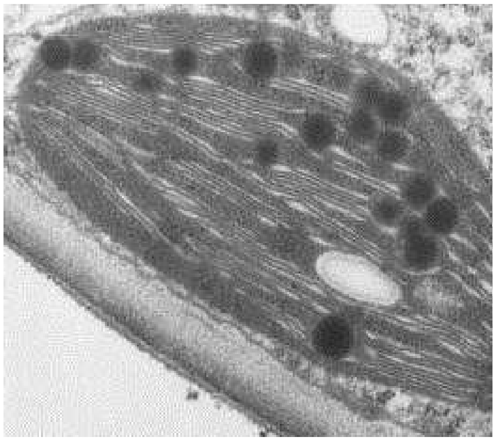
A Individual bacteria that are not affected by antibiotics evolve resistance.

B Bacteria that are exposed to methicillin will become resistant to all antibiotics to which they are exposed.

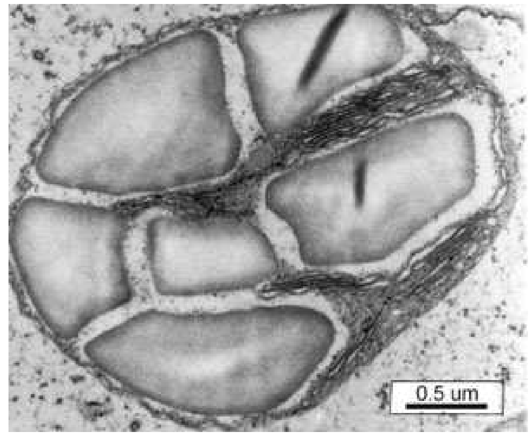
C A number of individuals in a population of bacteria may be resistant to methicillin.

D A person who is resistant to methicillin will pass on the methicillin-resistant gene to the bacteria.

23 The electron micrographs show two types of chloroplasts found in maize plants.



Chloroplast **P**



Chloroplast **Q**

Which of the following can be concluded from these photomicrographs of the maize chloroplasts?

1 There is intense photosynthetic activity in chloroplast **P** as compared to chloroplast **Q**.

2 GP, the final product of photosynthesis in chloroplast **P**, is transported to chloroplast **Q** where is it eventually converted to starch for storage.

3 Chloroplast **P** is found in regions of the maize plant which are exposed to more direct sunlight as compared to chloroplast **Q**.

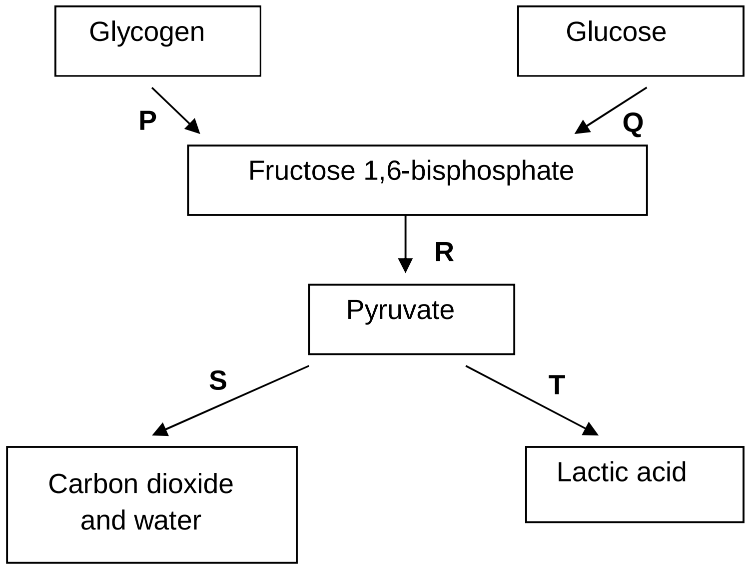
A 2 only

B 1 and 2 only

C 1 and 3 only

D All of the above

24 With reference to the diagram below, relate processes **P, Q, R, S, T** to statements **1, 2** and **3.**



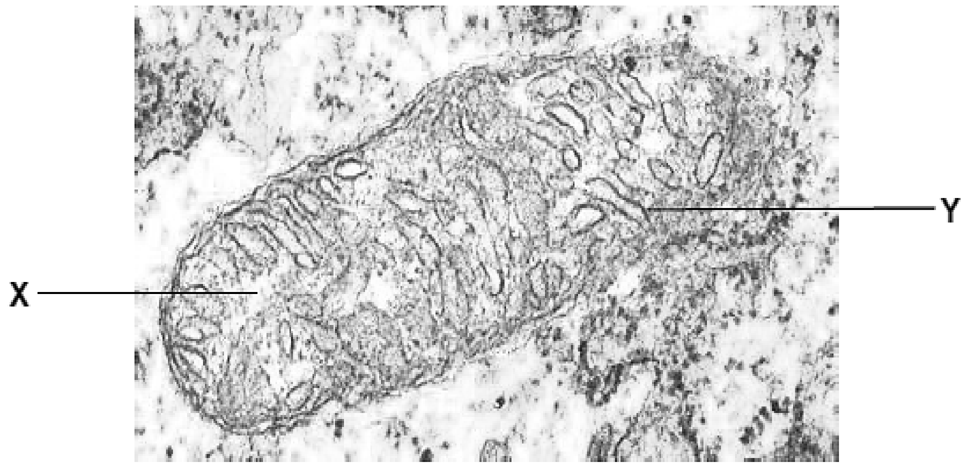
1 NAD^+ is regenerated without the use of the electron transport system.

2 ATP is synthesised via substrate level phosphorylation.

3 It can take place under anaerobic conditions.

	1	**2**	**3**
A	R	S	T
B	T	R	P
C	None of the above	S	R
D	S	Q	None of the above

25 The electron micrograph shows a mitochondrion with structures **X** and **Y** labelled.

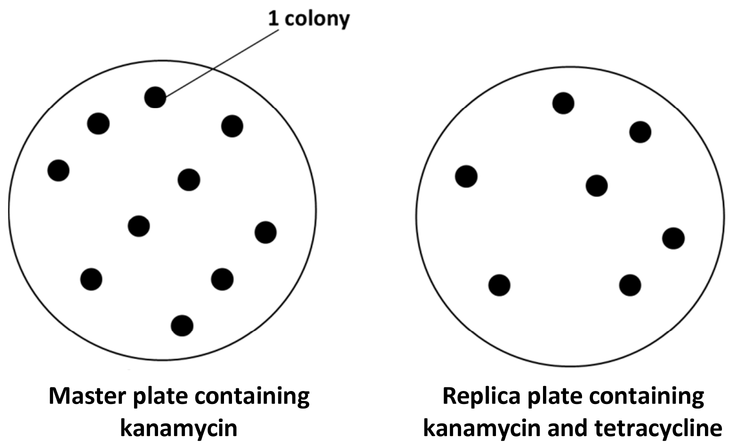


The phosphate : oxygen (P/O) ratio represents the number of ATP produced by oxidative phosphorylation per molecule of oxygen consumed. You may assume that the re-oxidation of each reduced coenzyme requires one oxygen molecule.

Under aerobic conditions, which of the following correctly represents the P/O ratio in structures **X** and **Y**?

	P/O ratio obtained by the complete oxidation of 1 molecule of glucose	
	Structure X	**Structure Y**
A	2.8	2.8
B	2.2	2.8
C	2.8	3.3
D	3.0	3.3

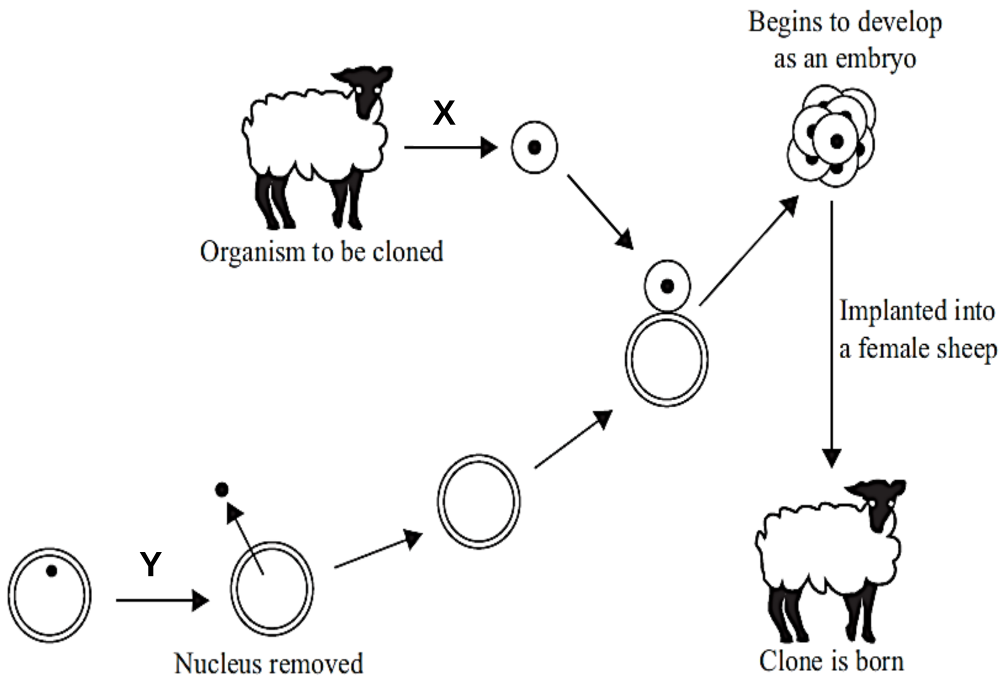
26 *E. coli* bacteria were transformed with plasmids containing a restriction site that cuts at one of its genetic markers. The bacteria were then plated onto nutrient agar plate containing kanamycin. Replica plating was subsequently carried out onto nutrient agar plate containing kanamycin and tetracycline. Bacterial growth on the two plates is shown below.



What conclusions about the position of the restriction site on plasmids and the bacterial colonies can be made?

	Restriction site on plasmids	**Master plate**	**Replica plate**
A	At kanamycin resistance gene	Consist of cells with recombinant & non-recombinant plasmid	Consist of cells with recombinant plasmids only
B	At kanamycin resistance gene	Consist of transformed cells	Consist of cells with recombinant plasmids only
C	At tetracycline resistance gene	Consist of cells with recombinant & non-recombinant plasmid	Consist of cells with non - recombinant plasmids only
D	At the tetracycline resistance gene	Consist of transformed cells	Consist of cells with recombinant plasmid cells

27 Which processes involved in cloning an animal are indicated by the letters **X** and **Y**?



	X	Y
A	Differentiated cell removed from animal	Nucleus removed from unfertilized egg cell
B	Sex cell removed from animal	Nucleus removed from differentiated animal cell
C	Sex cell removed from animal	Nucleus removed from unfertilized egg cell
D	Nucleus removed from unfertilized egg cell	Differentiated cell removed from animal

28 The human genome project has identified and mapped the genes on human chromosomes. This is allowing scientists to identify specific, faulty genes which contribute to inherited conditions.

This is useful in many ways, for example

1 Carriers of faulty genes can be identified and informed of their risk status.

2 Diagnostic tests can be developed to identify carriers of faulty genes.

3 Employers can take into account of the genetic predisposition of employees.

Which of these uses arise **directly** from the information provided by the project?

A 1 only

B 2 only

C 1 and 2

D 1 and 3

29 Which of the following statements is **not** a possible issue of concern over the creation of genetically modified farmed animals?

1 Genetic engineering may result in the creation of new proteins that are harmful to the organisms that produce or consume them.

2 Cross species gene transfer may compromise the genome integrity of the species involved.

3 Over production of certain gene products may cause undue stress to the genetically modified farmed animals.

4 Some genetically modified food products may not be acceptable to certain groups of people.

A 1 and 4

B 2 and 3

C 1, 3 and 4

D None of the above

30 It has been found that stem cells transferred from the intestinal lining to the bone marrow produce all of the different types of blood cell instead of intestinal cells.

Which statement explains this?

A All stem cells are totipotent.

B Environmental factors change the expression of specific genes.

C Specific genes are destroyed by endonucleases.

D Specific genes are hidden by condensation of some chromosomes.

~ End of Paper 1 ~

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